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CMPSC 463
04/26/2026

CrisisAid Route Planner Report

Description of the Project

The aim of the project is to design and deploy a web application to assist in the decision-making process during humanitarian crises. Users are able to select a starting point and a destination and the CrisisAid Route Planner algorithm will calculate the safest and most effective route between them. The program highlights destinations and represents routes with the help of an interactive map (Leaflet). Each location, including a community, shelter, or aid facility, is a location of a crisis area. Transit difficulty, risk, and distance are some of the factors that determine the cost of roads that are used to connect sites. This project will demonstrate how one could apply algorithmic solutions to address real-life problems, particularly when needed to make decisions quickly and under uncertain conditions.

Significance of the Project

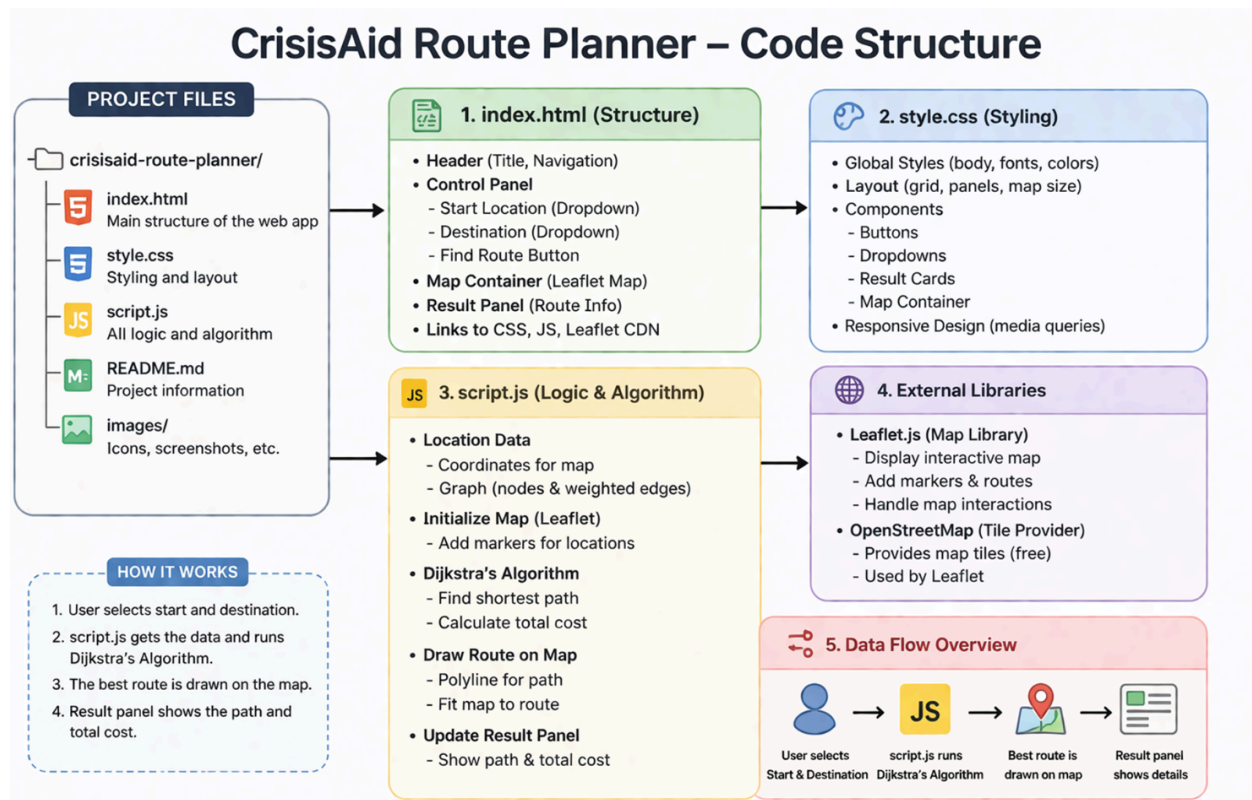
The project has significance in the sense that it shows how computer science may be applied to solve real humanitarian problems. In crises such as wars or natural disasters, delivering aid is often dangerous, time-sensitive, and unpredictable. Choosing an inefficient or unsafe route can delay assistance or increase risk for both aid workers and affected populations. This system demonstrates that it is possible to use an algorithm to analyze multiple possible paths and select the safest and the most effective one. Interactive map makes it more realistic and allows users to better comprehend the decision making process. Despite the simplified data that the system uses, it can well illustrate how real world systems can provide technology to support critical decision making processes.

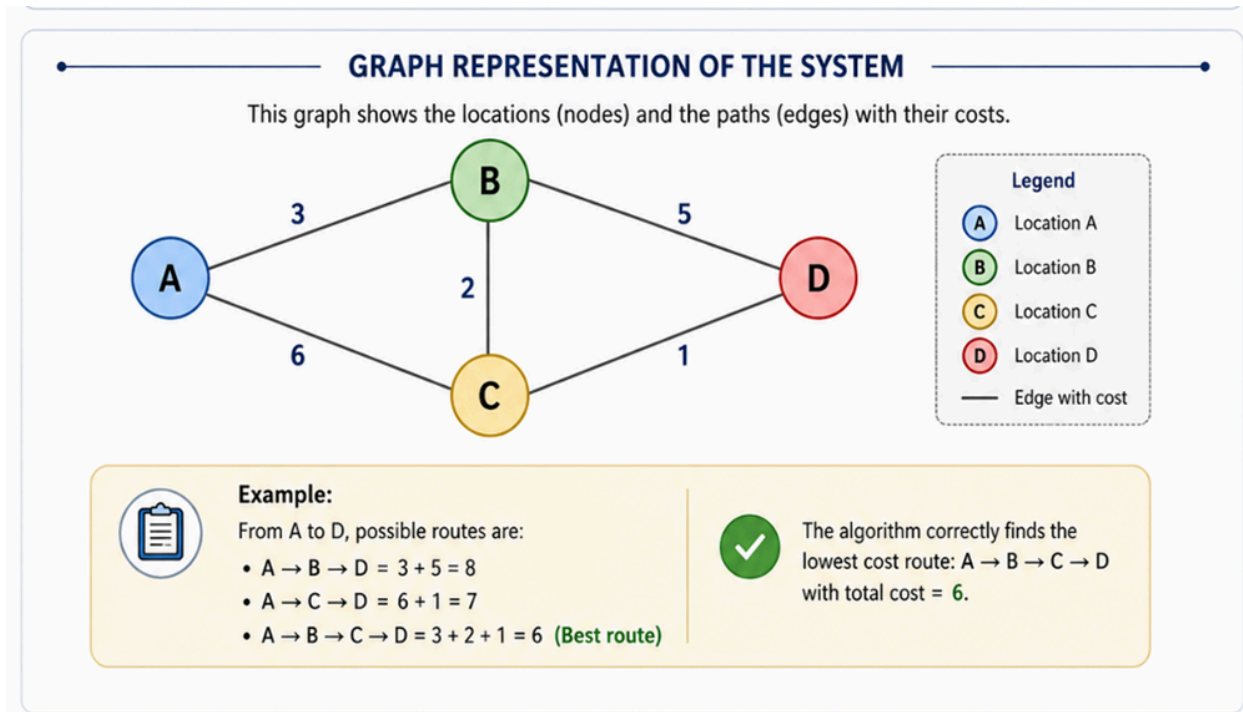
Code Structure

The project follows a clear and organized structure that separates the interface, design, and functionality. The index.html file is used to specify the layout of the application, which includes the header, user input controls, map container, and the output display section. CSS is used to do the styling, which regulates the design, colors, responsiveness and the general visual appeal of the application. The logic of the system is found in the JavaScript component. It captures location information, sets the graph structure, calculates the Dijkstra's Algorithm, sets up the map, sets markers at each point, draws the computed path on the map, and displays the findings to the user. This division of tasks enhances maintainability, scalability, and readability.

Code Structure Diagram

This structure explains how the diagram works, making the system easy to understand and maintain.





Description of Algorithms

The system uses the Algorithm of Dijkstra to determine the optimal route between two points. All locations in this project are considered to be nodes in a graph and all the routes between locations are considered to be edges with a cost. These costs represent a combination of travel challenges, safety, and distance. The algorithm begins with the selected starting point and assigns it an initial cost of 0. The unvisited node of the lowest known cost is then selected again and again and the cost of the neighboring node is updated. This process continues until the destination node is reached. The method that all costs the least is the safest and most effective.

Verification of Algorithms

To verify that the algorithm functions correctly, a simplified test scenario was used where multiple possible routes between locations were compared. The algorithm successfully identified the route with the lowest total cost, confirming that it correctly evaluates all available paths and selects the optimal one. By ensuring that the chosen route has the minimum cost among all possible options, the correctness of the implementation is validated.

Execution Results and Analysis

The application was tested with multiple starting and ending locations, and it consistently produced accurate results. The system uses Dijkstra's Algorithm to determine the best route when a user chooses a start and destination point. The result is displayed visually on the map

using a highlighted path and textually in the output section. The map automatically refines in order to work on the chosen route and it is easier to interpret the outcome by the users. It shows the overall effectiveness of using algorithm decision-making and visualization to increase usability and clarity. The system is still able to demonstrate how even more complex real-world data can be added in the future even with simplified data.

Conclusion

This project has been effective in showing how algorithmic methods can be used to the real-life problems of humanitarianism. The CrisisAid Route Planner employs Dijkstras Algorithm in cases of uncertainty and scarcity of resources in order to select the most efficient and safest route. The use of an interactive map enhances realism and understanding of the decision-making process by graphically presenting it. The system already operates on a simplified dataset, but it is a great base on which the improvements can be made in the future, i.e., integration of real-time data, updating the risk dynamically, and more powerful algorithms. In a synopsis, this project emphasizes the application of computer science concepts like graph algorithms and optimization to establish meaningful solutions that can be applied in making decisions during critical situations.

AI Usage

AI tools were used as a support resource throughout the development of this project. AI assisted in explaining algorithm concepts, debugging code, and helping generate the code structure diagram. AI helped me understand things better, but I still had to learn the material myself. It served as a tool to support development and improve the overall quality of the project.

GitHub

All components of the project, including the HTML structure, styling, JavaScript logic, and documentation, have been uploaded to a public GitHub repository. The repository is organized in a clear and accessible manner, allowing others to review, run, and evaluate the project. This ensures transparency and demonstrates proper project organization and version control.

Overall Quality of Report

This report demonstrates a clear understanding of algorithmic design and its application to real-world problems. The project is directly related to course concepts, including graph algorithms and optimization. All sections are well explained and the structure is in accordance with the guidelines. The writing is well structured and precise, showing good work in both execution and writing.